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HW 1 — Numerical simulation of the N-body problem

MAE 240 This script: (a) integrates the Newtonian N-body equations in first-order state-space form (b) plots 3D trajectories with initial/final markers (c) verifies center-of-mass motion (d) verifies angular momentum conservation (e) verifies total energy behavior

State vector: x = [r1; r2; ...; rN; v1; v2; ...; vN] in R^(6N)

Units used here: distance = km mass = kg time = s

```
clear; clc; close all;
```

1) Problem setup

```
G  = 6.67430e-20;        % km^3/(kg s^2)
AU = 1.496e8;            % km
yr = 365.25*24*3600;     % s

% ----------------------------
% Choose N >= 3 bodies
% ----------------------------
N = 4;

% Brief justification:
% One dominant central mass plus three lighter bodies gives a clear,
% stable test case for verifying center of mass, angular momentum,
% and total energy behavior.

bodyNames = {'Body 1','Body 2','Body 3','Body 4'};

% Masses [kg]
m = [1.989e30, 5.972e24, 6.39e23, 4.867e24];   % Sun/Earth/Mars/Venus-like
M = sum(m);

% Initial positions [km]
% Planar initial conditions: z = 0 for all bodies
r0 = zeros(N,3);
r0(1,:) = [0, 0, 0];
r0(2,:) = [1.000*AU, 0, 0];
r0(3,:) = [0, 1.524*AU, 0];
r0(4,:) = [-0.723*AU, 0, 0];
```

```
% Initial velocities [km/s]
% Approximately circular around Body 1
v0 = zeros(N,3);
v0(1,:) = [0, 0, 0];
v0(2,:) = [0,  sqrt(G*m(1)/norm(r0(2,:))), 0];
v0(3,:) = [-sqrt(G*m(1)/norm(r0(3,:))), 0, 0];
v0(4,:) = [0, -sqrt(G*m(1)/norm(r0(4,:))), 0];

% Optional: shift to exact zero total momentum frame
P0 = sum(m.' .* v0, 1);
vCM0 = P0 / M;
for i = 1:N
    v0(i,:) = v0(i,:) - vCM0;
end

% Pack into state vector x0 = [r_all; v_all]
x0 = [reshape(r0.',3*N,1); reshape(v0.',3*N,1)];

% Time span
t0   = 0;
tf   = 3*yr;
tout = linspace(t0, tf, 3000);

% ODE solver settings
opts = odeset('RelTol',1e-11,'AbsTol',1e-13);
```

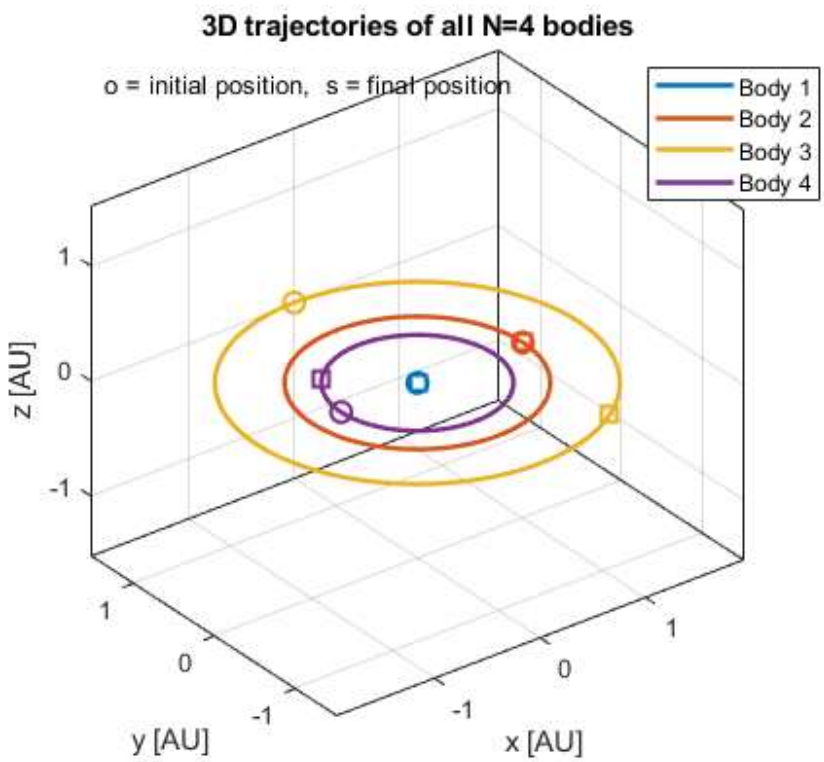
2) Integrate the first-order system xdot = f(x)

```
[T, X] = ode45(@(t,x) eom_nbody(t,x,m,G), tout, x0, opts);

% Extract position and velocity histories
Rhist = X(:,1:3*N);
Vhist = X(:,3*N+1:6*N);
```

3) Part (a): 3D trajectories

```
figure('Color','w');
hold on; grid on; box on; axis equal;

colors = lines(N);

% Store only orbit line handles for legend
hOrbit = gobjects(N,1);

for i = 1:N
    Ri = Rhist(:,3*i-2:3*i);

    % Orbit line
    hOrbit(i) = plot3(Ri(:,1)/AU, Ri(:,2)/AU, Ri(:,3)/AU, ...
        'LineWidth',1.8, 'Color',colors(i,:));

    % Initial marker
    plot3(Ri(1,1)/AU, Ri(1,2)/AU, Ri(1,3)/AU, ...
        'o', 'MarkerSize',8, 'LineWidth',1.5, ...
        'Color',colors(i,:), 'HandleVisibility','off');

    % Final marker
```

```
    plot3(Ri(end,1)/AU, Ri(end,2)/AU, Ri(end,3)/AU, ...
        's', 'MarkerSize',8, 'LineWidth',1.5, ...
        'Color',colors(i,:), 'HandleVisibility','off');
end

xlabel('x [AU]');
ylabel('y [AU]');
zlabel('z [AU]');
title('3D trajectories of all N=4 bodies');
legend(hOrbit, bodyNames, 'Location','best');
view(3);

% Note showing marker meaning
text(0.02, 0.95, 'o = initial position,  s = final position', ...
    'Units','normalized', 'FontSize',10);
```



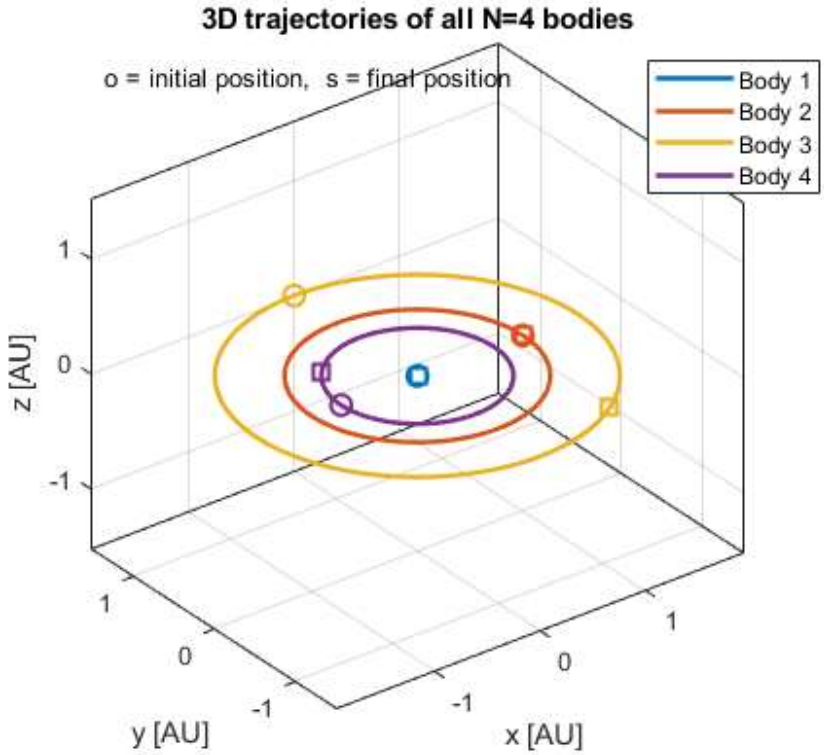
4) Part (b): Center of mass

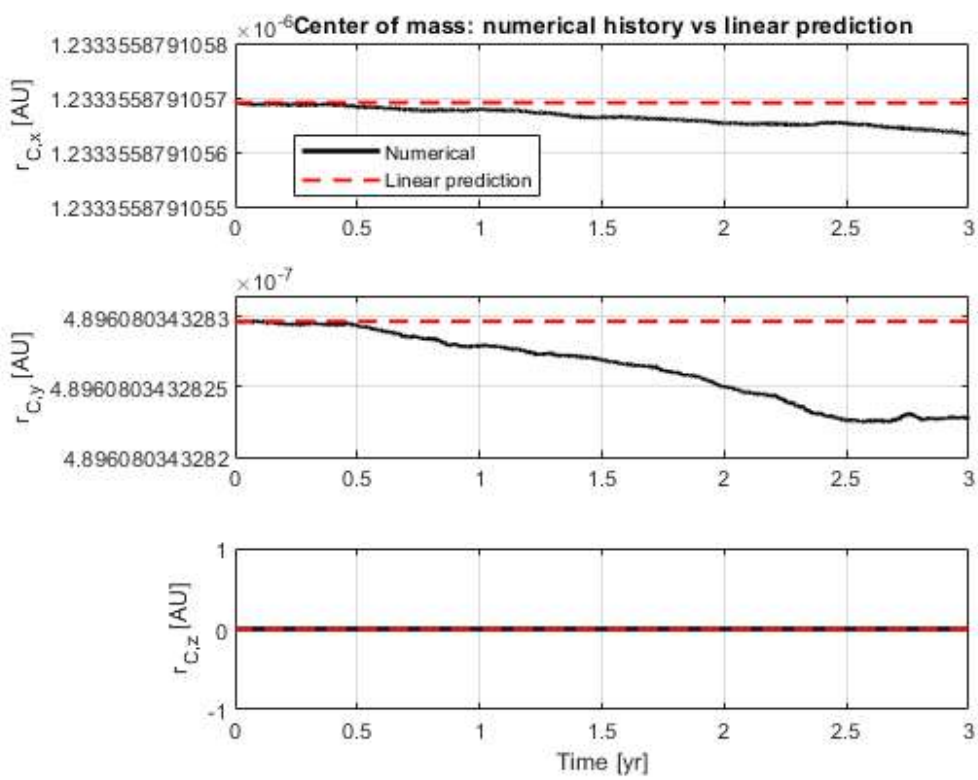
Compute numerical center of mass r_C(t)

```
rC = zeros(length(T),3);
vC = zeros(length(T),3);

for k = 1:length(T)
    rk = zeros(N,3);
    vk = zeros(N,3);

    for i = 1:N
        rk(i,:) = Rhist(k,3*i-2:3*i);
        vk(i,:) = Vhist(k,3*i-2:3*i);
    end
```

```
    rC(k,:) = (m * rk) / M;
    vC(k,:) = (m * vk) / M;
end

% Linear prediction:
% r_C(t) = r_C(t0) + rdot_C(t0)*(t-t0)
rC0 = rC(1,:);
vC0 = vC(1,:);
rC_pred = rC0 + (T - T(1))*vC0;

% Deviation
rC_err = vecnorm(rC - rC_pred, 2, 2);
max_rC_dev = max(rC_err);

fprintf('\n===== Center of Mass Check =====\n');
fprintf('Max deviation between numerical r_C(t) and linear prediction = %.6e km\n', max_rC_dev);

figure('Color','w');
tiledlayout(3,1);

labels = {'x','y','z'};
for j = 1:3
    nexttile;
    plot(T/yr, rC(:,j)/AU,'k','LineWidth',1.8); hold on;
    plot(T/yr, rC_pred(:,j)/AU,'r--','LineWidth',1.4);
    grid on;
    ylabel(sprintf('r_{C,%s} [AU]',labels{j}));
    if j == 1
        title('Center of mass: numerical history vs linear prediction');
        legend('Numerical','Linear prediction','Location','best');
    end
    if j == 3
        xlabel('Time [yr]');
    end
end
```

```
===== Center of Mass Check =====
Max deviation between numerical r_C(t) and linear prediction = 1.338799e-11 km
```





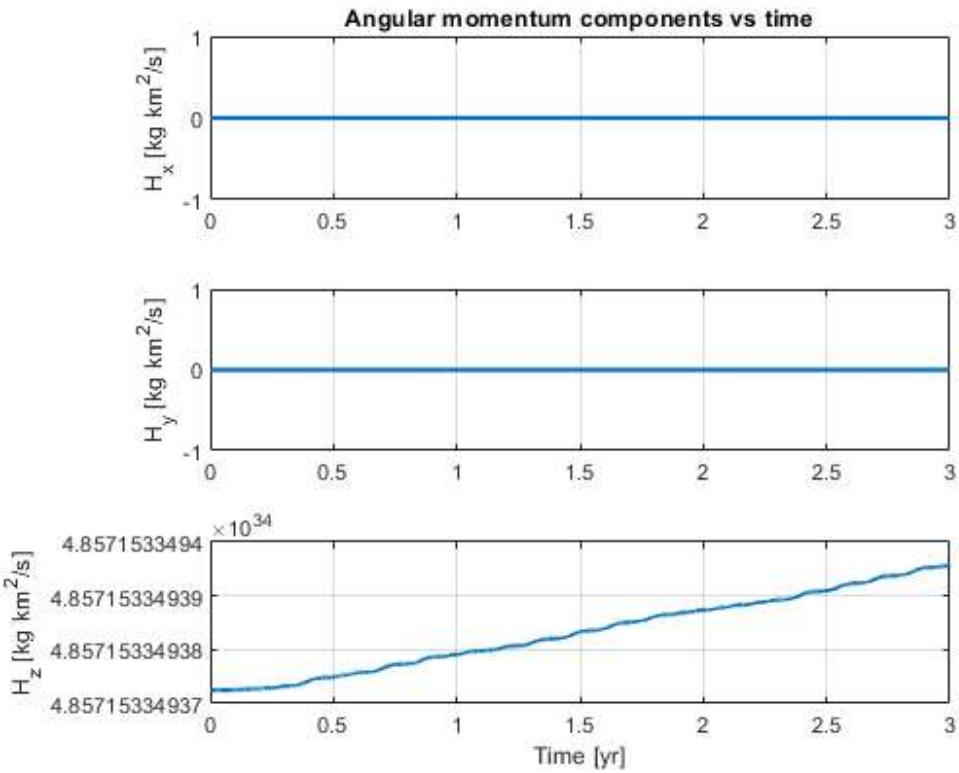
5) Part (c): Angular momentum

H(t) = sum_i m_i r_i x v_i

```
H = zeros(length(T),3);
```

```
for k = 1:length(T)
    Hk = [0 0 0];
    for i = 1:N
        ri = Rhist(k,3*i-2:3*i);
        vi = Vhist(k,3*i-2:3*i);
        Hk = Hk + m(i)*cross(ri,vi);
    end
    H(k,:) = Hk;
end

H0 = H(1,:);
H0_norm = norm(H0);
rel_H_drift = vecnorm(H - H0, 2, 2) / H0_norm;
max_rel_H_drift = max(rel_H_drift);

fprintf('\n===== Angular Momentum Check =====\n');
fprintf('Max relative drift ||H(t)-H(t0)|| / ||H(t0)|| = %.6e\n', max_rel_H_drift);

% Verify planarity
z_all = zeros(length(T),N);
for i = 1:N
    z_all(:,i) = Rhist(:,3*i);
end
max_abs_z = max(abs(z_all),[],'all');

fprintf('Max absolute out-of-plane displacement |z| = %.6e km\n', max_abs_z);

figure('Color','w');
tiledlayout(3,1);

for j = 1:3
    nexttile;
    plot(T/yr, H(:,j),'LineWidth',1.7);
    grid on;
    ylabel(sprintf('H_%s [kg km^2/s]',labels{j}));
    if j == 1
        title('Angular momentum components vs time');
    end
    if j == 3
        xlabel('Time [yr]');
    end
end
```

```
===== Angular Momentum Check =====
Max relative drift ||H(t)-H(t0)|| / ||H(t0)|| = 4.771240e-12
Max absolute out-of-plane displacement |z| = 0.000000e+00 km
```



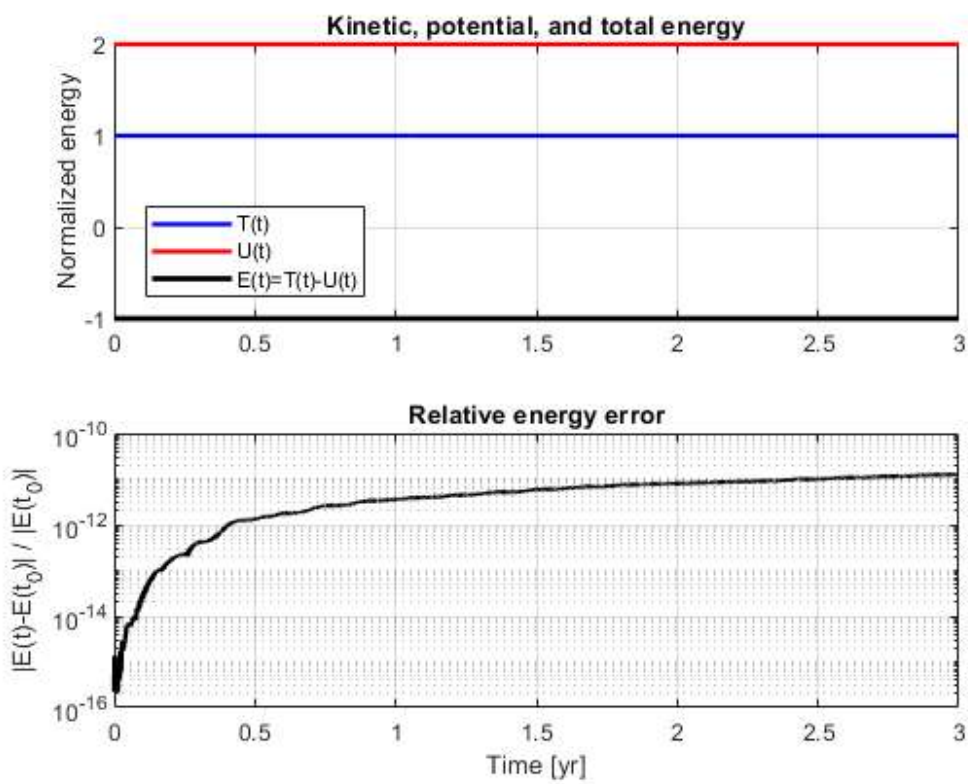
6) Part (d): Total energy

T = kinetic energy U = positive gravitational potential magnitude E = T - U

```
KE = zeros(length(T),1);
U  = zeros(length(T),1);
E  = zeros(length(T),1);

for k = 1:length(T)
    % Kinetic energy
    Tk = 0;
    for i = 1:N
        vi = Vhist(k,3*i-2:3*i);
        Tk = Tk + 0.5*m(i)*dot(vi,vi);
    end

    % Positive potential-energy magnitude U
    Uk = 0;
    for i = 1:N-1
        ri = Rhist(k,3*i-2:3*i);
        for j = i+1:N
            rj = Rhist(k,3*j-2:3*j);
            dij = norm(rj-ri);
            Uk = Uk + G*m(i)*m(j)/dij;
        end
    end

    KE(k) = Tk;
    U(k)  = Uk;
    E(k)  = Tk - Uk;
end

rel_E_err = abs(E - E(1)) / abs(E(1));
```

```
fprintf('\n===== Energy Check =====\n');
fprintf('Initial total energy E(t0) = %.6e kg km^2/s^2\n', E(1));
fprintf('Max relative energy error |E(t)-E(t0)| / |E(t0)| = %.6e\n', max(rel_E_err));

figure('Color','w');
tiledlayout(2,1);

nexttile;
plot(T/yr, KE/abs(E(1)), 'b','LineWidth',1.7); hold on;
plot(T/yr, U /abs(E(1)), 'r','LineWidth',1.7);
plot(T/yr, E /abs(E(1)), 'k','LineWidth',2.0);
grid on;
ylabel('Normalized energy');
title('Kinetic, potential, and total energy');
legend('T(t)','U(t)','E(t)=T(t)-U(t)','Location','best');

nexttile;
semilogy(T/yr, rel_E_err + eps,'k','LineWidth',1.7);
grid on;
xlabel('Time [yr]');
ylabel('|E(t)-E(t_0)| / |E(t_0)|');
title('Relative energy error');
```

```
===== Energy Check =====
Initial total energy E(t0) = -5.822528e+27 kg km^2/s^2
Max relative energy error |E(t)-E(t0)| / |E(t0)| = 1.252937e-11
```



7) Summary Section

```
fprintf('\n===== In summary =====\n');
fprintf('1. N = %d bodies were used.\n', N);
fprintf('2. The system was written in first-order state-space form xdot = f(x). \n');
fprintf('3. ode45 was used to integrate the equations of motion.\n');
fprintf('4. The center of mass followed the expected linear motion with max deviation %.3e km.\n', max_rC_dev);
fprintf('5. The angular momentum relative drift was %.3e.\n', max_rel_H_drift);
fprintf('6. The maximum energy relative error was %.3e.\n', max(rel_E_err));
fprintf(['7. The relative energy error stays very small during the entire simulation and increases \n' ...
    'only gradually over time. This behavior shows that the numerical integration is stable and \n' ...
    'accurately captures the system dynamics over the chosen time interval. The small drift is \n' ...
    'caused by the accumulation of truncation and roundoff errors from the ODE solver, which is \n' ...
    'expected when using a method such as ode45.\n']);
```

```
===== In summary =====
1. N = 4 bodies were used.
2. The system was written in first-order state-space form xdot = f(x).
3. ode45 was used to integrate the equations of motion.
4. The center of mass followed the expected linear motion with max deviation 1.339e-11 km.
5. The angular momentum relative drift was 4.771e-12.
6. The maximum energy relative error was 1.253e-11.
7. The relative energy error stays very small during the entire simulation and increases
only gradually over time. This behavior shows that the numerical integration is stable and
accurately captures the system dynamics over the chosen time interval. The small drift is
caused by the accumulation of truncation and roundoff errors from the ODE solver, which is
expected when using a method such as ode45.
```

==========

Local function ==========

```
function dxdt = eom_nbody(~, x, m, G)
    N = length(m);

    r = x(1:3*N);
    v = x(3*N+1:6*N);

    a = zeros(3*N,1);

    for i = 1:N
        ri = r(3*i-2:3*i);
        ai = [0;0;0];

        for j = 1:N
            if j ~= i
                rj = r(3*j-2:3*j);
                rij = ri - rj;
                dij = norm(rij);
                ai = ai - G*m(j)*rij/(dij^3);
            end
        end

        a(3*i-2:3*i) = ai;
    end

    dxdt = [v; a];
end
```

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